LECTURE 1

Introduction to Agents and Multiagent Systems

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1. Introduction

Artificial Intelligence (AI)

- **Symbolic AI**
  - symbolic representation of knowledge
  - logic, expert systems, rule based systems,…
- **Computational Intelligence** (CI)
  - numerical representation of knowledge
  - fuzzy systems, evolutionary computation, ML, **MAS**

Data mining (DM)

Machine Learning (ML)

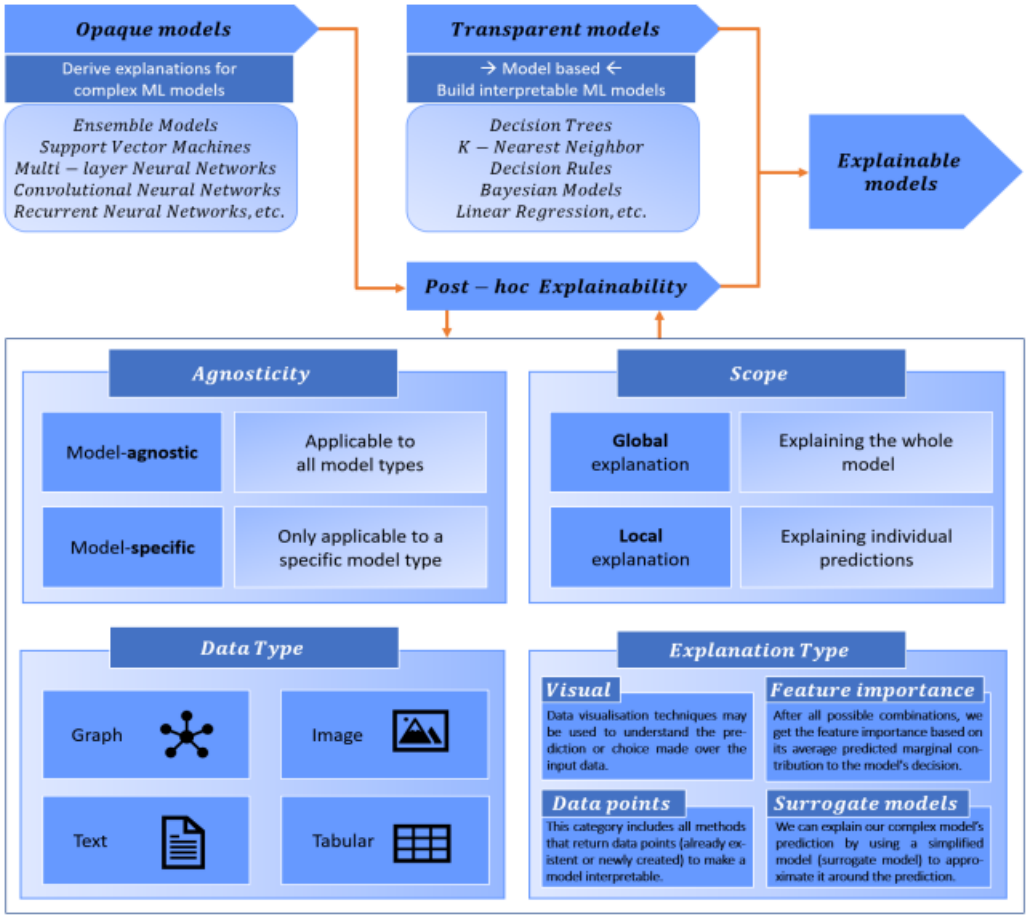
- Deep Learning (DL)

ML/DL

- **Generative AI/ML**
- Few shot learning
- Machine Unlearning
  - the process of selectively removing specific training data points and their influence on an already trained model, making the updated model behave the same as a model that was never trained on that data
- Representation/feature learning
  - Multimodal learning
    - Multimodal representation learning
  - Feature fusion
    - Computer vision tasks, bioinformatics, etc
- **Semi-supervised learning** (SSL taxonomy Yang et al, 2023)
- Contrastive learning
  - form of SSL
  - deep learning technique for unsupervised representation learning

AI

- interpretability of AI/ML/DL models
  - black-box vs white-box models
- Explainable AI (XAI)
  - understand and interpret the predictions of a ML model
- interpretability methods and evaluation of ML interpretability
  - Gradients, DeepLIFT, Class Activation Maps (CAMs), etc
- methods for ML interpretability can be classified according to various criteria



General ontology of XAI (link)

- **LIME** (**L**ocal **I**nterpretable **M**odel-agnostic **E**xplanations)
  - algorithm that can explain individual predictions of any classifier or regressor
- **SHAP** (**SH**apley **A**dditive ex**P**lanations)
  - a game theoretic approach to explain the output of any machine learning model

Trustworthy AI (TAI)

- not only **metrics**
- principle, tools
- ethics guidelines for TAI
- **other aspects**
  - robustness, algorithmic fairness, explainability, and transparency
  - robustness

- requirement of TAI
- refers to the degree that a model's performance changes when using new data versus training data
  - ideally, performance should not deviate significantly
- the ability of a model to maintain its performance when faced with uncertainties or adversarial conditions
  - noisy data, distribution shifts, and adversarial attacks
- how to test robustness?
  - cross-validation
  - adding noisy data
  - …

Application domains

- medicine
- bioinformatics
- education (*Educational Data Mining* - EDM)
- software engineering (*Search-based software engineering* - SBSE)
- psychology
- financial
- *e-business*: e-commerce, e-banking, e-gambling, etc
- computer vision
- meteorology
- ….

AI agents

- redefining productivity and innovation for businesses

Source: Boston Consulting Group, Inc.

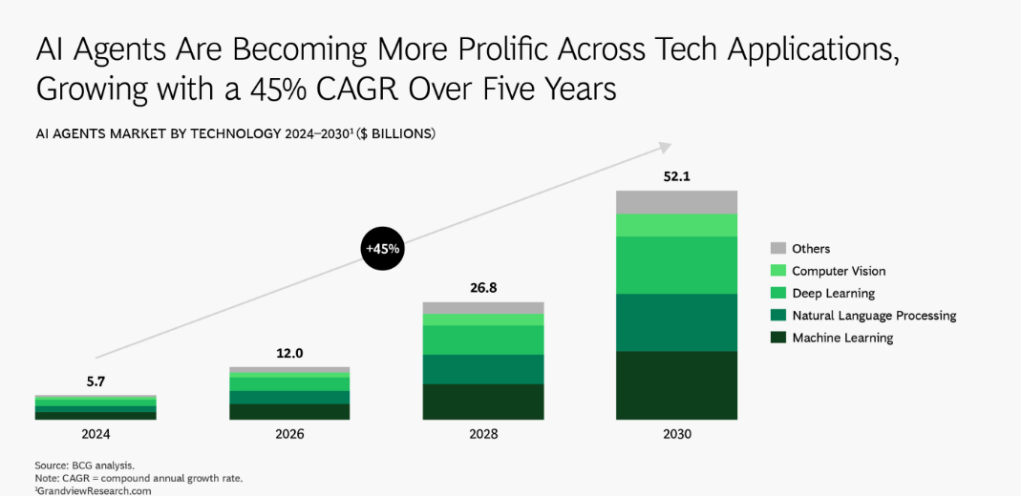
How Are Businesses Using AI Agents Today?

AI agents are fast becoming common across industries. Early adopters have already unlocked value from these intelligent software agents in multiple functions, including marketing and sales, customer service, R&D, and data and technology. But this is just the tip of the iceberg. Here are a few business cases for AI agents that companies are exploring today:

- **Marketing**: A leading consumer packaged goods company used intelligent agents to create blog posts, reducing costs by 95% and improving speed by 50x (publishing new blog posts in a single day as opposed to four weeks).
- **Customer service**: A leading global bank used AI virtual agents to interface with customers, reducing costs by 10x.
- **Research and development**: A biopharma company used AI agents for lead generation, reducing cycle time by 25% and gaining 35% in time efficiency for drafting clinical study reports.
- **Data and technology**: An IT department used AI agents to modernize its legacy technologies, increasing productivity by up to 40%.

Are AI Agents the Future?

AI agents are gaining traction quickly across an array of business applications—and the market for AI agents is expected to grow at a 45% CAGR over the next five years.



Agentic AI

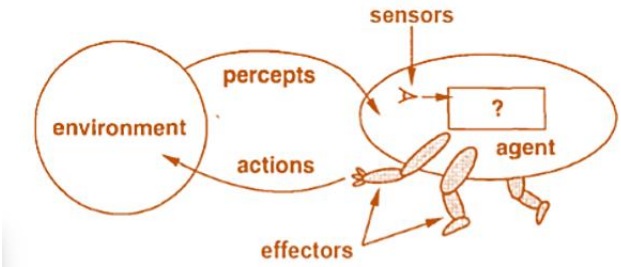
- **autonomous** systems (mainly acting based on LLMs) that can independently **plan, reason**, and **execute** multiple tasks with minimal human supervision
  - autonomous AI system that can act independently to achieve pre-determined goals
- software vendors (including Microsoft, Google, IBM) had adopted AI agents (since 2023)
  - embedding agentic AI capabilities directly in software platforms
- **AI agents** vs **agentic AI**
  - interchangeable?
  - differences?
  - communication, coordination, concurrency

2. Distributed Artificial Intelligence (DAI) & Multiagent Systems (MAS)

Distributed Artificial Intelligence (DAI)

– a research and application field which brings together and draws on results, concepts and ideas from many disciplines, including:
  - AI
  - SE
  - Sociology
  - Economics
  - Organization and management science
  - Philosophy

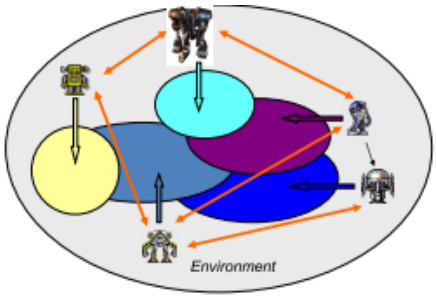
    - Game theory
    - Social sciences
    - …
- the study, construction and application of **multiagent systems (MAS)**, that is, systems in which several interacting (intelligent) agents pursue some set of goals or perform some set of tasks.
    - **agent**
        - model the concept of a human agent
        - computational entity such as a software program or a robot that can be viewed as perceiving and acting upon its environment
        - perception through **sensors**, actions through **effectors**/**actuators**
        - acts **autonomously** on behalf of their users



[https://www.educba.com/agents-in-artificial-intelligence/]

- **agents?**
    - **human agent**
        - eyes, ears, skin, etc. for **sensors**
        - hands, fingers, legs, mouth, etc. for **effectors**
        - powered by **muscles**
    - **robot**
        - camera, infrared, etc. for **sensors**
        - grippers, wheels, lights, speakers for **effectors**
        - often powered by motors
    - **software agent**
        - **functions** as **sensors**
            - information provided as input in the form of encoded bit strings or symbols
        - **functions** as **effectors**
            - results deliver the output

    - **MAS**
        - multiple agents in a common environment



- 1975-1980
- distributed (decentralized) solutions to AI problems
- directions in DAI
  - **DPS** (distributed problem solving)
    - field of DAI in which the objective lies in getting individual agents to work and interact with each other in order to solve problems that require a collective effort
  - traditionally there are two directions in DAI
    - **MAS**
      - in which several agents coordinate their knowledge and activities and reason about the processes of coordination.
      - initial work on behaviour coordination.
      - **toy problems** from the Agent/MAS domains
        - robotic football
        - predator and prey problem
        - Wumpus agent
        - Vacuum cleaning agent
    - **DPS**
      - the work of solving a particular problem is divided among a number of nodes that divide and share knowledge about the problem and the developing solution.
      - characteristics
        - speed up the solution process
        - expertise and problem solving may be distributed
        - the knowledge may be distributed
        - the solution may need to be distributed in order the solution to be executed
      - initial work on task decomposition and solution synthesis.
    - **DPS ↔ MAS**

Agents

- an important research and development area in the field of **Computer Science** and of **AI**
  - **AI/DAI**
    - agents – paradigm of AI
    - MAS – paradigm of DAI
  - **SE**
    - are a new paradigm for developing software applications
    - are related to other programming paradigms (e.g. OOP) → the **agent oriented programming paradigm** (AOP)
    - agent-oriented techniques represent powerful new means of analysis, designing and building complex software systems.
      - software abstraction
      - are considered a new important way in the conceptualization and implementation of different kind of applications
    - **agent based computing** has been hailed as a "new revolution in software"
- **issues**
  - **agents** vs. **intelligent** agents

        - **intelligence**
    - **agents** vs. programs
- **applications**
    - **industrial applications**
        - process control (robotics)
        - manufacturing (intelligent manufacturing, ERP, BPM)
        - air traffic control
    - **commercial applications**
        - information management
        - electronic commerce
        - business process management
            - viewing a BP as a society of negotiating, service providing agents
    - **medical applications**
        - patient monitoring
        - health care
    - **sciences**
        - geography (GIS – *geographical information systems*)
        - meteorology
            - weather forecasting, climatology, etc
    - **entertainment**
        - games
        - interactive theater and cinema
    - **other**
        - web mining, information retrieval, data mining, search engines, etc
- **agents** are frequently used nowadays in:
    - sociology, biology
    - cognitive psychology
    - social psychology
    - computer science
- **emotional agents**
    - affective computing

DAI – Agents

- the long term of **DAI** is to develop methods and mechanisms that enable **agents** to interact as well as humans (or even better) and to understand the interaction among intelligent entities whether they are computational, human or both
- the goal raises a number of challenging issues that are all centered around the elementary question of "WHEN and HOW to interact with WHOM"

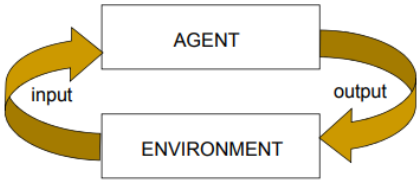
DAI – traditional AI

- both DAI and traditional AI deal with computational aspects of intelligence, but they do so from different points of view

Traditional AI	DAI
Cognitive processes within individuals	Social processes in groups of individuals
Concentrates on agents as "*intelligent stand-alone systems*" and on **intelligence** as a property of systems that act in isolation	Concentrates on agents as "*intelligent connected systems*" and on **intelligence** as a property of systems that interact
DAI is a generalization of traditional **AI**	

3. Agents

- continuous interaction with the environment
- sensor input, action output



An agent in its environment. The interaction between the agent and the environment is usually an ongoing, non-terminating one.

- One of the agent tasks/goals is to assist the user by making tasks for him, or teaching the user what he has to do
  - It has to successfully accomplish the task it was delegated to do

- **What an agent is?**
  - there is no such an agreement
  - there is no universally accepted definition of the term agent
  - there is a general consensus that **autonomy** is central to the notion of agency
  - An agent is a computer system that acts in behalf of a user, it is situated in some environment and that is capable of **autonomous** action in this environment, in order to meet its design objectives.
    1. **autonomy**
       - agents are able to act without intervention of humans or other systems, they have **control** both over their **internal state** and over their **behavior**.
- **Why agents?**
  - **delegation**
    - computers perform tasks for us, autonomously (without our intervention)
    - *give control* to computers
      - fly-by-wire aircrafts
      - fly-by-wire cars
      - intelligent braking systems
      - cruise control systems
      - etc

  - **programming progression**
    - programming has progressed

    machine code → assembly language → machine-independent programming languages → sub-routines → procedures and functions → ADTs → objects → **AGENTS**

  - **intelligence**
    - implement a rational/intelligent behavior

- **Delegation + Intelligence** ⇒ the need to build computer systems that can act effectively on our behalf to accomplish the tasks we delegated to them

- ability of computer systems to act *independently*
- the ability of computer systems to act in a way that represents our best interests while interacting with other humans/systems

- **Examples of trivial agents**
  - a **control system** (e.g. thermostat)
  - **software daemons**
    1. background processes in an operating system which monitor a software environment and perform actions to modify it
       - actions: software actions (e.g. changing an icon on the screen, executing a program, etc)
    2. e.g. **biff** command (UNIX)

4. Characteristics/properties of agents

- an **agent** is characterized by
  - an **architecture part**, named the behavior of the agent
    1. the action accomplished after a perceptual sequence
  - the **program part,** the internal part of the agent
    1. the task of AI is to develop the program part of the agent
    2. a function that implements the way to pass from perception to action
    3. the program part of an agent may be summarized as

Function AGENT-PROGRAM (*p*: Perception) **returns** an **Action**
  static: *memory*

  memory ← **UPDATE-MEMORY (***memory***,** *p*)
  a ← **CHOOSE-OPTIMAL-ACTION (***memory*)
  memory ← **UPDATE-MEMORY (***memory***,** *a*)

  returns *a*
EndFunction

Skeleton of an agent [2]

- **performance measure** (conceptually is outside the **agent**)
  - evaluates the success of the agent's performance

- an **agent** is capable of **flexible autonomous actions** to meet its design objectives
  - **flexibility**
    1. **reactivity:** the ability to perceive their environment and to respond to changes that occur in it
    2. **pro-activeness**: ability to exhibit a ***goal-directed behavior*** by taking the initiative in order to satisfy their design objectives
       - behavior not driven only by events, but capable of generating goals and acting **rationally** to achieve them
       - **rationality**
         - reasonable/acceptable decisions

            - agents select actions that make sense from the point of view of the information possessed by the agent and its goals (or the task for which it was designed)
        - in some situations (e.g., dynamic environments) **learning** is needed for preserving the agents' rationality
            - **learning** agents – **intelligent** agents
    - *rational agent* – selects actions to maximize the performance measure
    - *ideal rational agent*
        - for each possible percept sequence, such an agent selects the action that maximizes its performance measure
3. **social ability**
    - agents are capable of interacting with the agents (and possible humans) in order to meet their design objectives

5. Intelligent agents

- an **intelligent agent** must be endowed with:
    - an initial knowledge
    - the capability of **learning**
        - the **learning** capability assures the autonomy of the agent – the capability to deduce its behavior from its own experience
        - the agents that operate based only on their initial knowledge are not autonomous
            - they will act successfully only if their knowledge remains valid, loosing their **flexibility**
- an **intelligent agent** usually has a utility function, a measure of its actions' performance
    - u: $States \rightarrow \Re$
        - u associates to an environment state a real number (a measure of the state's success degree)
        - the agent prefers, at a given moment, the state having the maximum *utility*

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